

just list all the packages you want for your project in a text file and `pip` can install them all automatically. Python packages are diverse, with libraries that integrate with APIs, GUI programming frameworks, mathematical and biological computation, web frameworks and more.

If your designs are more mercenary, you'll be glad to know that there is healthy demand for Python programmers, and it has been placed at #8 or above in the TIOBE Programming Community Index from many years. It is used by major organisations like Google — where its creator worked till 2012 — and scientific research organisations such as CERN.

A popular video hosting website called YouTube is also built on Python, and it is used behind the scenes on other Google products as well. DropBox also uses Python, and in fact currently employs Guido. Another high-traffic website, Reddit is built on Python and is in fact open source. The popular Discus commenting platform is also built on Python, using the Django framework.

Many popular applications — 3ds MAX, Maya, GIMP among others — support Python for automating the application or for creating plugins.

Hello World

- `print("Hello, World")`

It's hard to make things any simpler than this, and Python is often about making things as simple as they can be.

Sort

- `def insertion_sort(ulist):`
- `for idx in range(1, len(ulist)):`
- `curr_value = ulist[idx]`
- `pos = idx`
- `while pos > 0 and ulist[pos - 1] > curr_value:`
- `ulist[pos] = ulist[pos - 1]`
- `pos -= 1`
- `ulist[pos] = curr_value`

Python code tends to be self-documenting with the tendency of many Python programmers to use longer variable and function names in Snake Case rather than CamelCase.

Of course, the above code need not be written in Python since it comes with an inbuilt function called `sort`. Python's `sort` happens to implement an excellent sorting algorithm that was invented for Python itself. It is called Timsort after its inventor Tim Peters and is widely used.